

Percentages & ratios

COMPLETE LEARNING GUIDE

What You'll Learn

Percentages and ratios are the most frequently reused concepts across the entire aptitude syllabus — profit/loss, data interpretation, mixtures, and even time-speed-distance problems are often percentage or ratio problems wearing a different costume. The single biggest scoring opportunity here is also the single biggest trap: successive percentage changes do not add linearly, and most students lose easy marks assuming a 20% rise followed by a 20% fall nets to zero when it actually nets to a 4% loss. This sub-module covers percentage-to-fraction fluency, the successive-change formula, the crucial distinction between percentage points and percentage change, ratio and proportion mechanics, combining ratios across a shared term, and the alligation method for mixture problems — all worked through with real numbers so the shortcuts are internalized, not just memorized as formulas to forget under exam pressure.

1. Percentage Basics and Fast Conversion

A percentage is just a fraction out of 100, and the fastest way to compute 'X% of Y' is to convert the percentage into its simplest fraction ($10\% = 1/10$, $12.5\% = 1/8$, $25\% = 1/4$, $33.33\% = 1/3$) and multiply — much faster than long multiplication with decimals, especially under time pressure.

- Find 15% of 340 without a calculator: $340 \times 15/100 = 340 \times 3/20 = 51$.

2. Successive Percentage Change — The Non-Additive Trap

When a value undergoes two percentage changes $a\%$ and $b\%$ in sequence, the net percentage change is NOT $(a+b)\%$. The correct formula is: $\text{net}\% = a + b + (ab/100)$, where a loss is entered as a negative value. This single formula prevents the most common percentage error in placement tests.

- 20% increase followed by a 20% decrease: $\text{net}\% = 20 + (-20) + (20 \times -20)/100 = 0 - 4 = -4\%$. Net result is a 4% DECREASE, not 0%.
- 10% increase followed by another 10% increase: $\text{net}\% = 10 + 10 + (10 \times 10)/100 = 20 + 1 = 21\%$, not 20%.

3. Percentage Points vs Percentage Change

A rate moving from 5% to 8% has risen by 3 percentage points, but the percentage increase (relative change) is $(3/5) \times 100 = 60\%$. News reports and exam questions frequently exploit this ambiguity, so always check whether a question asks for the absolute point difference or the relative percentage change.

- An interest rate rises from 5% to 8%: point increase = 3 percentage points; percentage increase = $3/5 \times 100 = 60\%$. These are very different numbers answering different questions.

4. Ratio and Proportion Fundamentals

A ratio $a:b:c$ divides a total into $(a+b+c)$ equal parts, and each share is (that term's ratio value) $\times$ (total $\div$ sum of ratio terms). Proportion problems ($a:b :: c:d$, so $ad=bc$) let you solve for an unknown term by cross-multiplication.

- Divide £750 among A, B, C in the ratio 2:3:5. Total parts = 10, so 1 part = £75. A = 150, B = 225, C = 375. Check: $150+225+375=750$.

5. Combining Two Ratios With a Common Term

When two ratios share a common variable (e.g., A:B and B:C), scale both ratios so the shared term matches (using the LCM of its two values), then merge them into a single three-term ratio.

- A:B = 2:3 and B:C = 4:5. LCM of B's values (3 and 4) is 12. Scale A:B = 2:3 to 8:12, and B:C = 4:5 to 12:15. Combined: A:B:C = 8:12:15.

6. Maintaining Constant Value: Inverse Percentage Adjustment

When price and quantity have an inverse relationship (expenditure = price $\times$ quantity must stay fixed), a price increase of $x\%$ requires a quantity reduction of $[x/(100+x)] \times 100$ percent to keep total expenditure unchanged — not a reduction of $x\%$ itself, which is a very common error.

- Sugar price rises 25%. To keep monthly sugar expenditure unchanged, reduce consumption by $[25/(100+25)] \times 100 = 20\%$, not 25%.

7. Alligation for Mixture Problems

The alligation rule finds the ratio in which two ingredients of different values must be mixed to get a desired average value: ratio = (higher value - mean value) : (mean value - lower value).

- Mix tea costing £40/kg with tea costing £60/kg to get a mixture costing £52/kg. Ratio = $(60 - 52) : (52 - 40) = 8:12 = 2:3$. So mix cheaper:costlier in ratio 3:2 (cheaper tea gets the larger share, since the mean is closer to the cheaper price).

Common Pitfalls

- Adding successive percentage changes directly (20% up then 20% down = 0%) instead of using $\text{net\%} = a + b + \frac{ab}{100}$ — the correct answer here is a 4% net loss.
- Confusing percentage points with percentage change — a rate moving from 5% to 8% is a 3-point rise but a 60% relative increase.
- Reducing consumption by the same percentage as a price increase to 'keep expenditure constant,' instead of using the inverse formula $\frac{x}{(100+x)} \times 100$.
- Combining two ratios without first scaling the shared term to a common value via LCM.
- Forgetting that a ratio must be simplified to lowest terms, or misreading 'divided in ratio' as 'divided equally.'
- In alligation, swapping which quantity gets the larger share — the ingredient closer in value to the mean gets the LARGER portion.

Key Takeaways

- Net successive % change = $a + b + \frac{ab}{100}$; never simply add two percentage changes.
- Percentage point difference " " percentage change — always check which one a question is asking for.
- To hold expenditure constant after a price change of $x\%$, adjust quantity by $\frac{x}{(100+x)} \times 100\%$, not by $x\%$ itself.
- Convert percentages to simple fractions (25%=1/4, 12.5%=1/8) for fast mental multiplication.
- Combine ratios by scaling the shared term to its LCM before merging.
- Alligation ratio = (higher " mean) : (mean " lower); the value closer to the mean gets the bigger share.

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